Jaesung Choi

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Employment

AI Fellow (AI Assistant Professor) Aug. 2026 – present
Korea Institute for Advanced Study (KIAS), Center for AI and Natural Sciences *Seoul, Korea*

AI Junior Fellow Oct. 2025 – Jul. 2026
Korea Institute for Advanced Study (KIAS), Center for AI and Natural Sciences *Seoul, Korea*
Mentor: Prof. Jaesung Lee

Visiting Scholar Feb. 2026 – May 2026
Arizona State University, School of Electrical, Computer and Energy Engineering *Tempe, AZ, USA*
Host: Prof. Ying-Cheng Lai (Regents Professor)

AI Research Fellow Mar. 2023 – Sep. 2025
Korea Institute for Advanced Study (KIAS), Center for AI and Natural Sciences *Seoul, Korea*
Mentor: Prof. Jaesung Lee

Education

Ph.D. in Mathematical Sciences Aug. 2019 – Feb. 2023
Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Korea*

- Advisor: Pilwon Kim, Ph.D., Professor in UNIST
- Dissertation: *Oscillator-based reservoir computing and its applications*

M.S. in Mathematical Sciences Feb. 2017 – Aug. 2019
Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Korea*

B.S. in Mathematical Sciences & Physics (double major) Mar. 2013 – Feb. 2017
Inha University *Incheon, Korea*

Research Interests

- **Machine Learning for Time Series:** Reservoir computing, autoencoder architectures, convolutional and attention mechanisms
- **Computational Methods:** Data augmentation and imputation algorithms, signal denoising, parameter extraction techniques
- **Nonlinear Dynamics:** Dynamical systems analysis, early warning indicators, critical transitions
- **Neuromorphic Hardware:** Physical reservoir computing implementations using Resistive Random Access Memory (RRAM) and Triboelectric Nanogenerators (TENG)

Publications

*(†equal contribution, *corresponding author)*

Journal Articles

2026

1. **J. Choi**†*, A.L. Chanu† and S. Awasthi†*. "Stochastic intracellular calcium dynamics show preserved structures identified by deep learning classification." *PLOS Computational Biology* 22.4 (2026): e1014240.

2. A.L. Chanu, Y. Park, **J. Choi**, Y. Cha, U. Lee, J. Moon and J. Park. "Human brain state classification via permutation entropy of EEG phase dynamics across consciousness levels and inattentive-type ADHD." *Chaos, Solitons & Fractals* 209 (2026): 118416.

2025

3. **J. Choi**, and P. Kim. "Unsupervised reservoir computing for multivariate denoising of severely contaminated signals." *Knowledge-Based Systems* (2025): 114574.

4. **J. Choi**, and P. Kim. "Homotopy Reservoir Computing: Harnessing chaos for computation." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35.9 (2025): 093112.

5. **J. Choi**, and P. Kim. "Signal-Noise separation using unsupervised reservoir computing." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35.8 (2025): 083136.

6. H. Park, D. Kim, I. Park, H. Nah, J. Yang, J. Yeo, **J. Choi**, J. Choi, H. Park and H. Choi. "Pd-modified microneedle array sensor integration with deep learning for predicting silica aerogel properties in real time." *ACS Applied Materials & Interfaces* (2025).

2023

7. **J. Choi**, E. Park, B. Jang and Y. Kim. "Reservoir concatenation and the spectrum distribution of concatenated reservoir state matrices." *AIP Advances* 13.11 (2023): 115110.

2022

8. **J. Choi**, and P. Kim. "Early warning for critical transitions using machine-based predictability." *AIMS Mathematics* 7.11 (2022): 20313-20327.

2020

9. **J. Choi**, and P. Kim. "Reservoir computing based on quenched chaos." *Chaos, Solitons & Fractals* 140 (2020): 110131.

2019

10. **J. Choi**, and P. Kim. "Critical neuromorphic computing based on explosive synchronization." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29.4 (2019): 043110.

Submitted Manuscripts

- **J. Choi**. "Predicting the onset of global chaos beyond the training range with topology-aware symplectic learning."
- **J. Choi**. "Extrapolating the emergence of Hamiltonian chaos with random-feature Hamiltonian neural networks." [arXiv:2607.28977]
- **J. Choi**, A.L. Chanu, and J. Park. "Learning a quantitative criterion for distinguishing chaos from noise."
- J. Park†, **J. Choi**†, H. Kim, H. Park, J. Kim, J. Choi, S. Kim, M. Lee, and K. Heo. "Reconstructing hidden variables in chaotic and real-world system with a single-memristor based physical reservoir computing."
- J. Park, H. Kim, J. Choi, **J. Choi*** and M. Lee*. "A triboelectric nanogenerator-driven physical reservoir computing for intelligent cardiac anomaly detection."

Manuscripts in Preparation

- **J. Choi**. "ForeSURE: Forecast-calibrated SURE for self-supervised broadband time-series denoising."
- **J. Choi**. "Thermodynamically admissible next-generation reservoir computing."

Presentations

Invited Talks

- "Reservoir Computing: Differences from Deep Learning, and Applications to State Reconstruction and Prediction." *Lecture and Tutorial* at the Institute for Future Earth, Pusan National University, 2026.
- "Learning chaos with artificial intelligence." *Special Lecture* at the Korea Academy of Complexity Studies–KENTECH Complex Systems Summer School, 2026. [poster]
- "Reservoir computing beyond the training regime." Seminar at the Nonlinear Dynamics & Mathematical Application Center (SRC), Kyungpook National University, 2026.
- "Reservoir computing and Echo state networks: Basics & Hands-on" *Tutorial* at 2026 UNIST Mathematical Imaging Lab Workshop, 2026.
- "Time series pattern discovery and harnessing complex systems for computation." Seminar at KIAS 2025 Winter Research Exchange Meeting, 2025.
- "Reservoir computing: From chaotic system prediction to physical implementation." Seminar at Inha University, 2025.
- "Extracting predictable information from time series data using machine learning." Seminar at National Institute for Mathematical Sciences (NIMS), 2025.
- "Extracting predictable information from time series data using machine learning." 1st APCTP-SISSA Joint Workshop on AI and Theoretical Physics, 2024.
- "Multivariate signal-noise separation using unsupervised reservoir computing." KSIAM 2024 Annual Meeting, 2024.
- "How reservoir computing differs from conventional neural networks and its potential applications." Seminar at National Institute for Mathematical Sciences (NIMS), 2024.
- "Reservoir computing: Concepts and Implementations." *Tutorial* at 2024 KPS Fall Meeting - Semiconductor Physics Department, 2024.
- "Strengths and recent research trends of reservoir computing." Seminar at Korea University, 2024.
- "Harnessing dynamical systems for computation - The reservoir computing approach." Seminar at Kyungpook National University, 2024.
- "Introduction to physical reservoir computing." Seminar at Hanyang University (ERICA), Inha University, 2024.
- "Brief introduction to physical reservoir computing." Non-Equilibrium Statmech Theory (NEST) Meeting at KIAS, 2024.
- Review on the paper "Reservoir computing with error correction: Long-term behaviors of Stochastic Dynamical Systems." 11th Workshop on Nonequilibrium Fluctuation Theorems, 2023.
- "Constructing Efficient Reservoir Computers and Exploring Memory Capacity." Seminar at Korea Institute of Science and Technology, 2022.

Contributed Talks

- Review on the paper "Reservoir-computing based associative memory and itinerancy for complex dynamical attractors." NEST Meeting at KIAS, 2024.
- "Exploring the computational potential of Titanium Dioxide Thin Films as physical reservoir computer." KIAS Center for AI and Natural Sciences Workshop, 2024.
- "Denoising extremely noisy dynamical systems using unsupervised reservoir computing." KSIAM 2024 Spring Conference, 2024.
- "Signal-Noise separation using unsupervised Echo state networks." KIAS Center for AI and Natural Sciences Workshop, 2023.

- "Early Warning Indicators for noisy systems." UNIST ML-based Early Warning Indicators Workshop, 2023.

Grants

KIAS Individual Grant (AP092902) Mar. 2025 – Feb. 2028

- Principal Investigator (PI), Center for AI and Natural Sciences, KIAS
- Research Subject: Predictive modeling research based on interaction between time series pattern discovery and neural network dynamical structures
- Amount: 13 million KRW per year

KIAS Individual Grant (AP092901) Mar. 2023 – Feb. 2025

- Principal Investigator (PI), Center for AI and Natural Sciences, KIAS
- Research Subject: Realization of reservoir computing framework based on complex systems
- Amount: 10 million KRW per year

Honors and Awards

UNIST Graduate Full Scholarship Mar. 2017 – Feb. 2023

Service and Outreach

Alternative Military Service

Technical Research Personnel Mar. 2023 – Feb. 2025

- Affiliation: Korea Institute for Advanced Study (KIAS)

Technical Research Personnel Mar. 2022 – Feb. 2023

- Affiliation: Ulsan National Institute of Science and Technology (UNIST)

Referee Services

Manuscripts reviewed for: Communications AI & Computing, Expert Systems With Applications (ESWA), Journal of the Korean Physical Society (JKPS), PLOS ONE, Proceedings of the National Academy of Sciences (PNAS), Review of Scientific Instruments (RSI), Scientific Reports.

Career Talks

- "Postdoctoral research experience in applied mathematics for undergraduate and graduate students." Math Night at UNIST, 2025.
- "Experiences as a researcher with a degree in mathematics." Seminar in Career Exploration Class at Kyung-Hee University, 2024.

Other Services

- Organizer and Session Chair, 2026 Winter Workshop on PDE and Applied Mathematics
- Session Chair, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2026
- Task Force Member, KIAS-Lite (Small Language Model) Development Project, Korea Institute for Advanced Study, 2025
- Session Chair, KIAS Lecture Series on Computational Neuroscience and AI, 2025
- Organizer, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2025
- Session Chair, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2024
- Organizer, UNIST Machine Learning-based Early Warning Indicator Workshop, 2023